

Origami Polyhedra

Hands-on Mathematics, IdeaLab VITM

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A regular tetrahedron is made of four equilateral triangle faces.

Task 1:

To fold an equilateral triangle from a square piece of paper.

Task 2:

Making a tetrahedron using triangle units.

Problem 1:

Verify Euler's formula $V - E + F = 2$.

Problem 2:

Number the vertices, faces, and edges of the tetrahedron 1 to 14 in such a way that the number on each edge is the mean of the two numbers at the vertices joined by that edge and also the mean of the two numbers on the faces it separates?

Task 3:

Making an octahedron using triangle units.

Task 4:

Making an icosahedron using triangle units.

Task 5:

Making a cube using Sonobe units.